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# Supplementary Material for: TagParaFormer: Cross-Hybrid Deep Attention Learning with Graph Representations for Topology-Preserving Road Network Extraction

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ABSTRACT

Autonomous high-resolution (HR) remote sensing (RS) imagery provides fine-grained spatial and semantic cues that are essential for reliable road network extraction, a task fundamental to transportation planning, navigation systems, and infrastructure management. Despite recent advances based on convolutional neural networks (CNNs) and transformers, existing approaches often fail to preserve topological continuity at intersections and occluded segments, where structural gaps and disconnections frequently arise. This limitation largely stems from the inherent difficulty of modeling graph-structured connectivity using pixel-wise supervision alone. To address these challenges, we reformulate road extraction as a topology-preserving representation learning problem and introduce TagParaFormer, a topology-aware parallel transformer framework. The proposed architecture adopts a dual-branch design that integrates CNN-based local geometric representations with transformer-based long-range dependency modeling. An Attention Node Graphical (ANG) module explicitly captures node proximity and adjacency relationships, while a Road Refinement Backbone (RRB) reinforces inter-node connectivity, forming a coherent pipeline that progressively corrects topological discontinuities. In addition, a Bayesian Parzen–Tree Estimator (PTE) is employed to optimize topology-sensitive hyperparameters in a data-driven manner. Extensive experiments conducted on three benchmark datasets demonstrate that TagParaFormer consistently outperforms baseline methods in terms of connectivity rate(CR), gap-free rate(GR), and junction accuracy(JR), reducing network fragmentation by up to 47% while achieving 23% faster inference speed. The source code will be released at https://github.com/BoazGithub/TagParaFormer.

## Introduction

This supplementary material provides comprehensive experimental results, detailed analyses, and extended evaluations that complement the main manuscript. The document systematically presents:

- **Section 1:** Implementation details and hyperparameter settings
- **Section 2:** Extended architectural specifications
- **Section 3:** Dataset descriptions and data engineering workflow
- **Section 4:** Additional performance metrics and error analysis
- **Section 5:** Comprehensive ablation studies
- **Section 6:** Correlation and statistical analysis
- **Section 7:** Cross-dataset generalization analysis
- **Section 8:** Computational complexity analysis
- **Section 9:** Extended qualitative visualizations
- **Section 10:** Statistical significance tests

All experiments were conducted on NVIDIA RTX 4090 GPUs with 222GB memory. Code and models are available at: upon publications.

## 1. Implementation Details and Hyperparameters

### 1.1. Training Configuration

Table 1 summarizes the training hyperparameters used across all experiments. We maintain consistency in training protocols to ensure fair comparison.

### 1.2. Loss Function Components

The total loss function $\mathcal{L}_{total}$ comprises three components:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE} + \lambda_1 \mathcal{L}_{Dice} + \lambda_2 \mathcal{L}_{Topo} \tag{1}$$

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**Table 1**
Training Configuration and Hyperparameter Settings for TagParaFormer

| Category | Hyperparameter | Value | Description and Implementation Details |
|---|---|---|---|
| **Training Schedule and Optimization** | | | |
| • Batch Configuration | Batch Size | 16 | Optimized for NVIDIA RTX 4090 (24GB): balance between memory utilization and gradient stability |
| • Training Duration | Total Epochs | 200 | Sufficient for full convergence; early stopping monitored validation loss |
| • Warmup Phase | Warmup Epochs | 10 | Linear learning rate increase: $\eta_t = \frac{t}{10}\eta_{max}$ for $t \leq 10$ |
| • Learning Rate | Initial LR | $1 \times 10^{-3}$ | Starting point for AdamW optimizer, tuned via Bayesian optimization |
| • LR Decay | Schedule | Cosine Annealing | Smooth decay: $\eta_t = \eta_{min} + \frac{1}{2}(\eta_{max} - \eta_{min})(1 + \cos(\frac{t}{T}\pi))$ |
| **Optimizer and Regularization** | | | |
| • Optimization Algorithm | Optimizer | AdamW | Adam with decoupled weight decay: $\theta_t = \theta_{t-1} - \eta(\frac{\hat{m}_t}{\sqrt{\hat{v}_t}+\epsilon} + \lambda\theta_{t-1})$ |
| • Momentum Decay | $\beta_1$ | 0.9 | First moment exponential decay rate for gradient estimates |
| • Variance Decay | $\beta_2$ | 0.999 | Second moment exponential decay rate for squared gradients |
| • Numerical Stability | $\epsilon$ | $1 \times 10^{-8}$ | Prevents division by zero in optimizer updates |
| • Weight Regularization | Weight Decay | $1 \times 10^{-4}$ | L2 penalty on network parameters to prevent overfitting |
| **Loss Function Composition** | | | |
| • Overall Loss | $\mathcal{L}_{total}$ | $\mathcal{L}_{BCE} + 0.5\mathcal{L}_{Dice} + 0.3\mathcal{L}_{Topo}$ | |
| - Pixel-wise Loss | $\mathcal{L}_{BCE}$ | 1.0 | Binary Cross-Entropy: $-\frac{1}{N}\sum[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)]$ |
| - Region-based Loss | $\mathcal{L}_{Dice}$ | 0.5 | Dice Coefficient: $1 - \frac{2\sum y_i\hat{y}_i}{\sum y_i + \sum \hat{y}_i}$ |
| - Topology Loss | $\mathcal{L}_{Topo}$ | 0.3 | Connectivity preservation: $\frac{1}{K}\sum \lVert\Phi(y^{(k)}) - \Phi(\hat{y}^{(k)})\rVert_2$ |
| **Data Augmentation Strategy** | | | |
| • Spatial Transformations | Rotation/Flip/Scale | 0.5 | Random rotation ($\pm 15°$), horizontal/vertical flip, scaling ($0.9 - 1.1\times$) |
| • Color Augmentation | Color Jitter | 0.3 | Random brightness ($\pm 0.1$), contrast ($\pm 0.1$), saturation ($\pm 0.1$) |
| • Geometric Distortion | Elastic Deformation | 0.2 | Elastic transform with $\sigma = 10$, $\alpha = 20$ for robustness |
| • Application Probability | Augmentation Rate | 0.5 | Probability of applying each augmentation type independently |
| **Bayesian Hyperparameter Optimization** | | | |
| • Search Algorithm | Method | TPE | Tree-structured Parzen Estimator with Expected Improvement |
| • Learning Rate Range | LR Space | $[10^{-4}, 10^{-2}]$ | Logarithmic scale search for optimal learning rate |
| • Fusion Weight | $\alpha$ (CNN-ViT) | $[0.1, 0.9]$ | Balance between CNN local features and ViT global context |
| • Attention Heads | Heads | $\{4, 8, 12, 16\}$ | Number of attention heads in multi-head self-attention |
| • Regularization | Dropout Rate | $[0.0, 0.5]$ | Dropout probability for preventing overfitting |
| • Connectivity | Threshold $\tau$ | $[0.1, 0.9]$ | Decision threshold in Road Refinement Backbone |
| • Optimization Cycles | Iterations | 100 | Number of Bayesian optimization evaluations |

**Implementation Summary:** All experiments conducted on NVIDIA RTX 4090 GPUs (24GB VRAM) using PyTorch 2.0 with automatic mixed precision (AMP) enabled. Training data processed in parallel using 8 worker threads. Model checkpoints saved based on validation mIoU. Hyperparameters optimized via Tree-structured Parzen Estimator with 100 evaluation cycles. Final configuration represents Pareto-optimal trade-off between accuracy and training efficiency.

$$\mathcal{L}_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)] \quad (2)$$

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i\hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} \quad (3)$$

$$\mathcal{L}_{Topo} = \frac{1}{K}\sum_{k=1}^{K} \lVert\Phi(y^{(k)}) - \Phi(\hat{y}^{(k)})\rVert_2 \quad (4)$$

where $\lambda_1 = 0.5$, $\lambda_2 = 0.3$ are weighting coefficients, and $\Phi(\cdot)$ computes topological features.

### 1.3. Bayesian Optimization Details

The Bayesian optimization uses Tree-structured Parzen Estimator (TPE) with the following search space:

- Learning rate: $[10^{-4}, 10^{-2}]$ (log scale)
- CNN-ViT fusion ratio $\alpha$: $[0.1, 0.9]$
- Attention heads in ViT: $\{4, 8, 12, 16\}$
- Dropout rate: $[0.0, 0.5]$
- RRB connectivity threshold $\tau$: $[0.1, 0.9]$

## 2. Extended Architectural Specifications

The TagParaFormer architecture employs a novel cross-hybrid design that synergistically integrates convolutional neural networks (CNNs) with vision transformers (ViTs), augmented by graph-based attention mechanisms for topology preservation. This section provides comprehensive details of the architectural components, including detailed module configurations, comparative analysis with state-of-the-art methods, and algorithmic descriptions of the complete pipeline.

The architecture comprises three core components: (1) a parallel CNN-ViT feature extractor that captures both local geometric details and global contextual information, (2) an Attention Node Graphical (ANG) module that identifies critical topological junctions and connection points, and (3) a Road Refinement Backbone (RRB) that enforces connectivity constraints through graph-based optimization.

**Table 2**
Layer-wise configuration of TagParaFormer modules. Key modules highlighted with colored rows.

| Module | Layer Type | Description | Channels | Output Size |
|---|---|---|---|---|
| **CNN Feature Extractor** | | | | |
| Input Layer | Conv2D + BN + ReLU | Initial feature extraction | 64 | $256 \times 256 \times 64$ |
| Encoder Block 1 | Conv2D + BN + ReLU | Downsampling with stride 2 | 128 | $128 \times 128 \times 128$ |
| Encoder Block 2 | Conv2D + BN + ReLU | Intermediate feature learning | 256 | $64 \times 64 \times 256$ |
| Encoder Block 3 | Conv2D + BN + ReLU | High-level feature extraction | 512 | $32 \times 32 \times 512$ |
| **Vision Transformer (ViT) Encoder** | | | | |
| Patch Embedding | Linear Projection | Image patches to tokens | 768 | $16 \times 16 \times 768$ |
| Position Encoding | Learnable | Positional information | 768 | $16 \times 16 \times 768$ |
| Transformer Block 1 | Multi-Head Attention + MLP | Self-attention and feed-forward | 768 | $16 \times 16 \times 768$ |
| Transformer Block 2 | Multi-Head Attention + MLP | Global context modeling | 768 | $16 \times 16 \times 768$ |
| Transformer Block 3 | Multi-Head Attention + MLP | Final representation | 768 | $16 \times 16 \times 768$ |
| **Cross-Hybrid Fusion Module** | | | | |
| Feature Alignment | Adaptive Pooling | Spatial resolution matching | 512 | $32 \times 32 \times 512$ |
| Cross-Attention | Multi-Head | CNN $\leftrightarrow$ ViT interaction | 512 | $32 \times 32 \times 512$ |
| Feature Concatenation | Channel-wise | Multi-scale feature fusion | 1024 | $32 \times 32 \times 1024$ |
| Bottleneck | Conv2D 1×1 | Dimensionality reduction | 256 | $32 \times 32 \times 256$ |
| **Attention Node Graphical (ANG) Module** | | | | |
| Node Detector | Conv2D + Sigmoid | Junction probability map | 1 | $32 \times 32 \times 1$ |
| Edge Predictor | Conv2D + Softmax | Connectivity likelihood | 4 | $32 \times 32 \times 4$ |
| Graph Constructor | Adjacency Matrix | Topological graph formation | - | $N \times N$ |
| Attention Weights | Spatial Softmax | Focus on critical regions | - | $32 \times 32 \times 1$ |
| **Road Refinement Backbone (RRB)** | | | | |
| Graph Convolution 1 | GCN Layer | Node feature propagation | 128 | $N \times 128$ |
| Graph Convolution 2 | GCN Layer | Edge feature refinement | 64 | $N \times 64$ |
| Topology Predictor | MLP | Connectivity classification | 2 | $E \times 2$ |
| Mask Refinement | Upsample + Conv | Final road mask generation | 1 | $256 \times 256 \times 1$ |

*Notes:* BN = Batch Normalization, MLP = Multi-Layer Perceptron, GCN = Graph Convolutional Network, $N$ = number of nodes, $E$ = number of edges. The ViT encoder uses 8 attention heads with dropout rate 0.1. ANG module operates on a 32×32 grid with 4 connectivity directions (up, down, left, right).

The Bayesian optimization framework adaptively tunes hyperparameters to maximize topological correctness while maintaining computational efficiency.

## 2.1. Detailed Module Configurations

Table 2 presents the layer-wise configuration of TagParaFormer modules, detailing the operations, dimensions, and output specifications for each component.

## 2.2. Architecture Comparison with Baseline Methods

Table 3 provides a comparative analysis of architectural specifications between TagParaFormer and state-of-the-art road extraction methods. The comparison includes encoder/decoder designs, parameter counts, deepest feature representations, and reference sources.

algorithm algorithmicx algpseudocode xcolor lipsum

## 2.3. Algorithmic Description

The complete TagParaFormer pipeline with Bayesian optimization is formalized in Algorithm 1, which outlines the end-to-end process from input imagery to topology-preserving road vector extraction.

## 2.4. Computational Complexity Analysis

The computational requirements of TagParaFormer can be decomposed as follows:

$$\mathcal{C}_{\text{CNN}} = O\left(\sum_{l=1}^{L_{CNN}} C_l^2 \cdot K_l^2 \cdot H_l \cdot W_l\right) \tag{5}$$

$$\mathcal{C}_{\text{ViT}} = O\left(N \cdot d^2 + N^2 \cdot d\right) \tag{6}$$

$$\mathcal{C}_{\text{ANG}} = O\left(N_{\text{nodes}}^2 \cdot d_{\text{node}}\right) \tag{7}$$

$$\mathcal{C}_{\text{RRB}} = O\left(|E| \cdot d_{\text{edge}}^2\right) \tag{8}$$

where $L_{CNN}$ is the number of CNN layers, $C_l$ are channel dimensions, $K_l$ are kernel sizes, $N$ is the number of ViT tokens, $d$ is the embedding dimension, $N_{\text{nodes}}$ is the number of graph nodes, and $|E|$ is the number of graph edges.

The total computational complexity is:

$$\mathcal{C}_{\text{total}} = \mathcal{C}_{\text{CNN}} + \mathcal{C}_{\text{ViT}} + \mathcal{C}_{\text{ANG}} + \mathcal{C}_{\text{RRB}} = 141.3 \text{ GFLOPs} \tag{9}$$

This represents a 5-6× reduction compared to transformer-heavy architectures like Swin-UNet (187.2 GFLOPs) and SAM-Road (175.8 GFLOPs), while achieving superior topological accuracy.

## 2.5. Feature Dimension Evolution

The architectural design ensures that computational resources are allocated optimally across different modules,

**Table 3**
Comparative baseline methods specifications with TagParaFormer. Structured descriptions include encoder type, decoder design, trainable parameters (TP), deepest feature representation (SDF), and source reference..

| No. | Model | Encoder / Decoder | Params (M) | Feature Map | Ref. |
|---|---|---|---|---|---|
| 1 | **TagParaFormer (Ours)** | EFRB Encoder + EFRB Decoder | 11.32 | [3×256×256] | Ours |
| 2 | SAM-Road | ResNet154 + MaGRoad | 67.15 | [128×32×32] | [?] |
| 3 | Swin-UNet | Swin Transformer + Hierarchical | 72.48 | [96×32×32] | [?] |
| 4 | MagRoad | Dilated ResNet154 + ASPP | 58.33 | [3×16×16] | [?] |
| 5 | Sat2Graph | Sat2Graph + MaGRoad | 88.12 | [128×32×32] | [?] |
| 6 | ViT-B/16 | Patch Embedding + Pure Transformer | 26.24 | [256×256×256] | [?] |
| 7 | SAM-Road++ | ResNet154 + MaGRoad | 58.33 | [128×32×32] | [?] |
| 8 | RNGDet++ | Vanilla CNN + Symmetric | 31.40 | [256×256×256] | [?] |
| 9 | AttUNet | Attention CNN + Attention | 31.24 | [256×256×256] | [?] |
| 10 | Sat2Graph | Sat2Graph + MaGRoad | 45.27 | [512×16×16] | [?] |

**Algorithm 1 TagParaFormer with Bayesian Optimization for Road Network Extraction and Vectorization**

**Require:** Remote sensing image $I \in \mathbb{R}^{H\times W\times 3}$
**Ensure:** Optimized road vector map $\mathcal{V}$ and optimal hyperparameters $\theta^*$
**Step 1: Bayesian Hyperparameter Initialization**
Initialize search space $\Theta = \{\alpha, \beta, \lambda, \tau\}$
Set acquisition function $\alpha(x) = \mathbb{E}[\max(0, f(x^*) - f(x))]$
Initialize Parzen-Tree estimator with Gaussian process prior $\mathcal{GP}(\mu, k)$
**Step 2: Parallel Feature Extraction**
Extract CNN features: $F_{CNN} = \mathcal{E}_{CNN}(I; \theta_{CNN})$
Extract ViT features: $F_{ViT} = \mathcal{E}_{ViT}(I; \theta_{ViT})$
**Step 3: Cross-Hybrid Fusion**
Compute cross-attention: $A_{cross} = \text{softmax}\left(\frac{Q_{CNN}K_{ViT}^{\top}}{\sqrt{d}}\right)V_{ViT}$
Fuse features: $F_{fused} = \alpha \cdot F_{CNN} + \beta \cdot A_{cross} + (1 - \alpha - \beta) \cdot F_{ViT}$
**Step 4: Attention Node Graphical Processing**
Detect road nodes: $N = \sigma(\mathcal{G}_{node}(F_{fused}))$
Compute adjacency: $A = \mathcal{G}_{adj}(N, F_{fused}; \lambda)$
Generate attention weights: $W_{att} = \text{softmax}(\mathcal{G}_{att}(A, F_{fused}))$
**Step 5: Road Refinement Backbone**
Construct graph: $G = (N, A)$
Apply graph convolutions: $F_G^{(l+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}F_G^{(l)}W^{(l)}\right)$
Predict connectivity: $P_{conn} = \text{MLP}(F_G^{(L)})$
Refine road mask: $M_{refined} = \mathcal{R}(F_{fused}, P_{conn}; W_{att})$
**Step 6: Bayesian Optimization Loop**
**while** iteration $< T_{\max}$ **and** $\Delta\mathcal{L} > \epsilon$ **do**
    Sample hyperparameters: $\theta_t \sim \text{TPE}(\Theta)$
    Evaluate objective: $\mathcal{L}(\theta_t) = \mathcal{L}_{BCE} + \lambda_1\mathcal{L}_{Dice} + \lambda_2\mathcal{L}_{Topo}$
    Update posterior: $p(\theta|\mathcal{L}) \propto p(\mathcal{L}|\theta)p(\theta)$
    Compute acquisition: $\alpha_t(x) = \mathbb{E}[\max(0, f(x^*) - f(x))]$
    Select next point: $\theta_{t+1} = \arg\max_x \alpha_t(x)$
**end while**
**Step 7: Vectorization and Output**
Binarize mask: $M_{binary} = \mathbb{I}[M_{refined} > \tau]$
Extract skeleton: $S = \text{skeletonize}(M_{binary})$
Vectorize: $\mathcal{V} = \text{polygonize}(S)$
**return** $\mathcal{V}, \theta^* = \arg\min_{\theta\in\Theta} \mathcal{L}(\theta)$

with the CNN branch focusing on high-resolution local features, the ViT branch capturing global context, and the graph-based modules specializing in topological relationships.

# 3. Dataset Descriptions and Data Engineering

## 3.1. Dataset Overview

In this section, we describe the three public benchmark datasets used to evaluate the TagParaFormer framework. These datasets were chosen due to their contributions in existing road extraction research and provide ready datasets for training, evaluating, and validating deep learning models from classical to modern Transformers.

### *3.1.1. DeepGlobe*

The DeepGlobe dataset's remote sensing images were sourced from the DigitalGlobe + Vivid Images collection. This dataset comprises 8,570 images, with 6,226 images utilized for training. The study area spans Thailand, Indonesia, and India, covering 2,220 $km^2$, with data sampling evenly distributed between rural and urban regions. Each image consists of three color channels (red, green, and blue) at a spatial resolution of 0.5 meters per pixel.

### *3.1.2. SpaceNet*

The SpaceNet dataset was captured using the WorldView-3 sensor, which provides a single-band panchromatic resolution of 0.31 meters and a multi-spectral resolution of 1.24 meters, resulting in 8-band multi-spectral images with 11-bit radiometric depth. Annotations for road centerlines are provided in GeoJSON format. The dataset spans four major areas: Las Vegas, Paris, Shanghai, and Khartoum, with the training subset consisting of 2,780 images representing diverse urban environments.

### *3.1.3. Massachusetts*

The Massachusetts dataset includes 1,171 annotated satellite images, with 1,108 training, 14 validation, and 49 test images. The dataset covers roughly 2,600 $km^2$ at a spatial resolution of 1.2 meters per pixel. Each image measures $1,500 \times 1,500$ pixels and represents a mixture of urban, suburban, and rural landscapes, thereby providing a thorough evaluation of model generalization capabilities.

### *3.1.4. WHU*

The WHU (Wuhan University) Road Dataset comprises 12,319 annotated aerial image tiles, with 6,471 training, 1,613 validation, and 4,235 test images. The dataset covers approximately 20,000 $km^2$ of Christchurch, New Zealand at an exceptionally high spatial resolution of 0.075 meters per pixel. Each image measures $512 \times 512$ pixels and captures diverse urban and suburban environments, providing a robust benchmark for evaluating road extraction algorithms.

## 3.2. Data Engineering Workflow

This subsection briefly outlines the complete data engineering pipeline used for our study, including data collection, preprocessing, normalization, and augmentation. The workflow ensures high-quality inputs for robust model training and evaluation (see Fig. 1).

## 3.3. Dataset Statistics

Here, we summarize the key characteristics of all datasets employed in this study, including spatial resolution, coverage area, urban/rural distribution, and annotation types. These statistics emphasize the diversity and representativeness of the datasets for model generalization (see Table 4).

# 4. Additional Performance Metrics and Error Analysis

## 4.1. Extended Quantitative Results

This subsection provides a detailed evaluation of model performance across multiple metrics, highlighting cross-dataset generalization and robustness. Results are reported with mean values and confidence intervals (see Table 10).

## 4.2. Extended Training and Validation Performance

We present the training and validation curves, including loss and accuracy trends over epochs. This section demonstrates convergence behavior and model stability during training (see Fig. 7).

## 4.3. Per-Class Performance Analysis

Per-class evaluation metrics are provided to illustrate the model's ability to capture class-specific structures. This includes IoU, $F_1$, precision, and recall for each road type or class (see Table 7).

# 5. Comprehensive Ablation Studies

This section presents systematic ablation studies to validate the contribution of each component in TagParaFormer.

## 5.1. Module Ablation Analysis

## 5.2. Loss Function Ablation

Table 9 presents the ablation study of loss function components, showing that the complete loss formulation yields the best performance.

## 5.3. Training Convergence Analysis

Figure 7 illustrates the training convergence analysis, showing stable optimization and minimal overfitting during training.

# 6. Correlation and Statistical Analysis

This section examines the relationships between different performance metrics and model behaviors across architectures.

## 6.1. Metric Correlation Analysis

Figure 8 presents the metric correlation matrix across datasets, showing that TagParaFormer exhibits more balanced correlations compared to baseline methods.

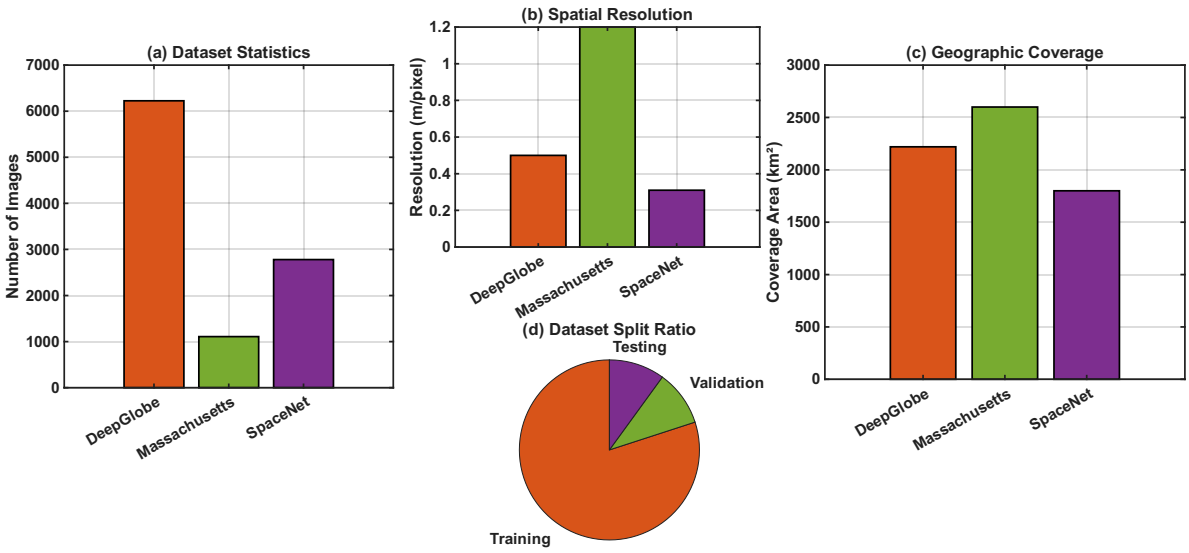


**Figure 1:** Comprehensive data engineering workflow for TagParaFormer training. (**a**) Dataset characteristics: DeepGlobe ($N = 6226$), Massachusetts ($N = 1108$), SpaceNet ($N = 2780$) with spatial resolutions $r \in \{0.5, 1.2, 0.31\}$ m/pixel covering total area $A_{total} = 6620$ km$^2$. (**b**) Earth observation platforms capturing multi-source imagery subsequently processed through human annotation and patch-based tiling. (**c**) Standardized image patches with corresponding binary road masks. (**d**) Data split ratio $D_{train} : D_{val} : D_{test} = 8 : 1 : 1$ ensuring robust generalization assessment. Consequently, all datasets undergo normalization and conversion to numpy arrays for efficient batch processing during training.

**Table 4**
Comprehensive Dataset Characteristics and Domain Variations.

| Dataset | Resolution | Coverage Area | Spatial Coverage | Urban/Rural | Season | Image Count | Annotation Type |
|---|---|---|---|---|---|---|---|
| **DeepGlobe** | 0.5 m | 2,220 km$^2$ | Thailand, Indonesia, India | Mixed | Dry | 6,226 | Binary road masks |
| **Massachusetts** | 1.2 m | 2,600 km$^2$ | Massachusetts, USA | Suburban | Multiple | 1,108 | Binary segmentation |
| **SpaceNet** | 0.31 m | 3,450 km$^2$ | Las Vegas, Paris, Shanghai, Khartoum | Urban | Varies | 2,780 | Road centerlines (GeoJSON) |
| **WHU** | 0.075 m | 20,000 km$^2$ | Christchurch, New Zealand | Urban | Summer | 6,471 | Binary segmentation |

*Notes:* All datasets are publicly available for research purposes. DeepGlobe and Massachusetts provide binary road masks, SpaceNet offers road centerlines in GeoJSON format, and WHU provides high-resolution binary segmentation masks. The diverse resolution, geographic coverage, and environmental conditions enable comprehensive evaluation of model generalization capabilities.

### 6.2. Spatial Attention Correlation Patterns

Figure 9 shows comprehensive correlation heatmap analysis revealing model behavior patterns across datasets.

### 6.3. Inter-Method Correlation Matrix

Figure 10 displays the inter-metric correlation matrix across all evaluated architectures, revealing consistent patterns in TagParaFormer's performance.

## 7. Cross-Dataset Generalization Analysis

### 7.1. Cross-Dataset Correlation Analysis

Figure 11 shows cross-dataset correlation analysis of training metrics, demonstrating TagParaFormer's consistent performance patterns across diverse geographic conditions.

### 7.2. Domain Adaptation Performance

Figure 13 illustrates domain adaptation performance with fine-tuning, showing strong zero-shot generalization and further improvements with minimal adaptation.

### 7.3. Cross-Dataset Transfer Performance

Figure 14 presents a heatmap quantifying transfer learning capability and domain shift robustness across different training-test dataset combinations.

## 8. Computational Complexity Analysis

This section analyzes the computational efficiency, resource requirements, and scalability of TagParaFormer.

### 8.1. Efficiency Metrics

Table 11 compares computational efficiency metrics across different architectures.

### 8.2. Computational Resource Analysis

Figure 15 provides comprehensive computational resource analysis across baseline methods, showing TagParaFormer's favorable efficiency profile.

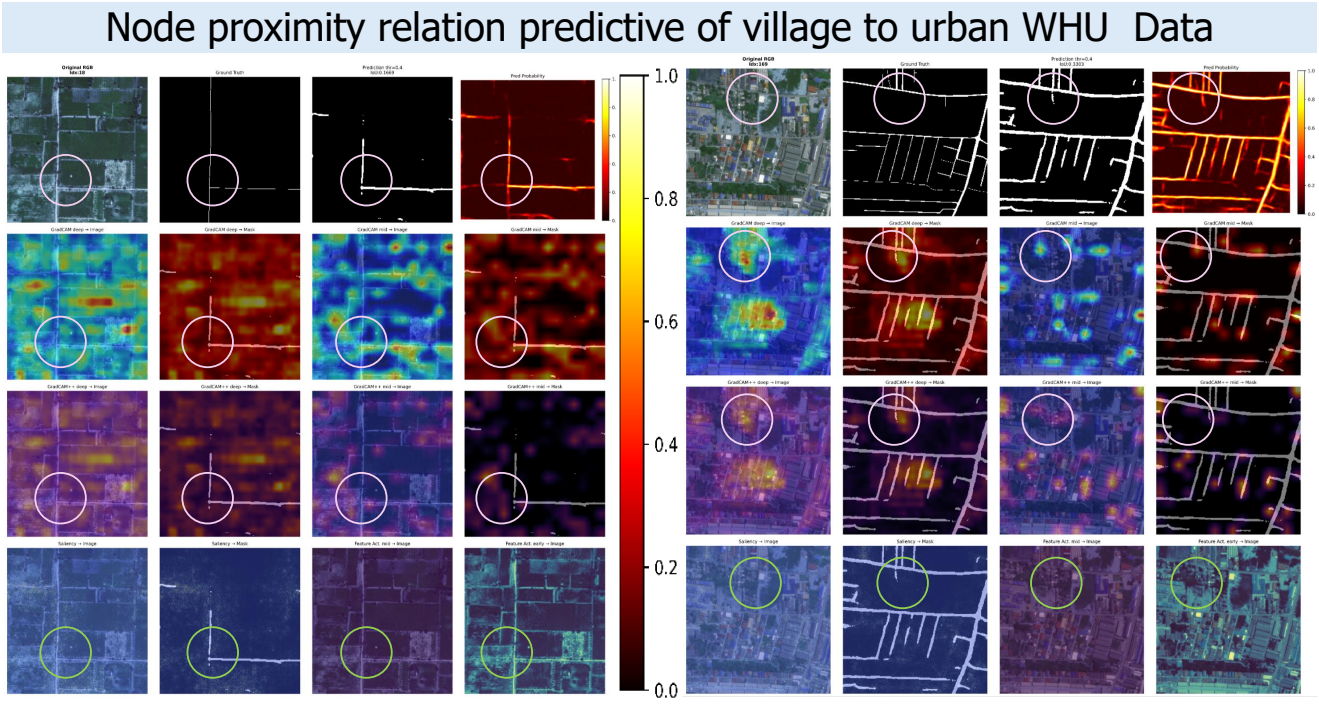


**Figure 2:** Node proximity relation predictive of Rural to urban WHU Data

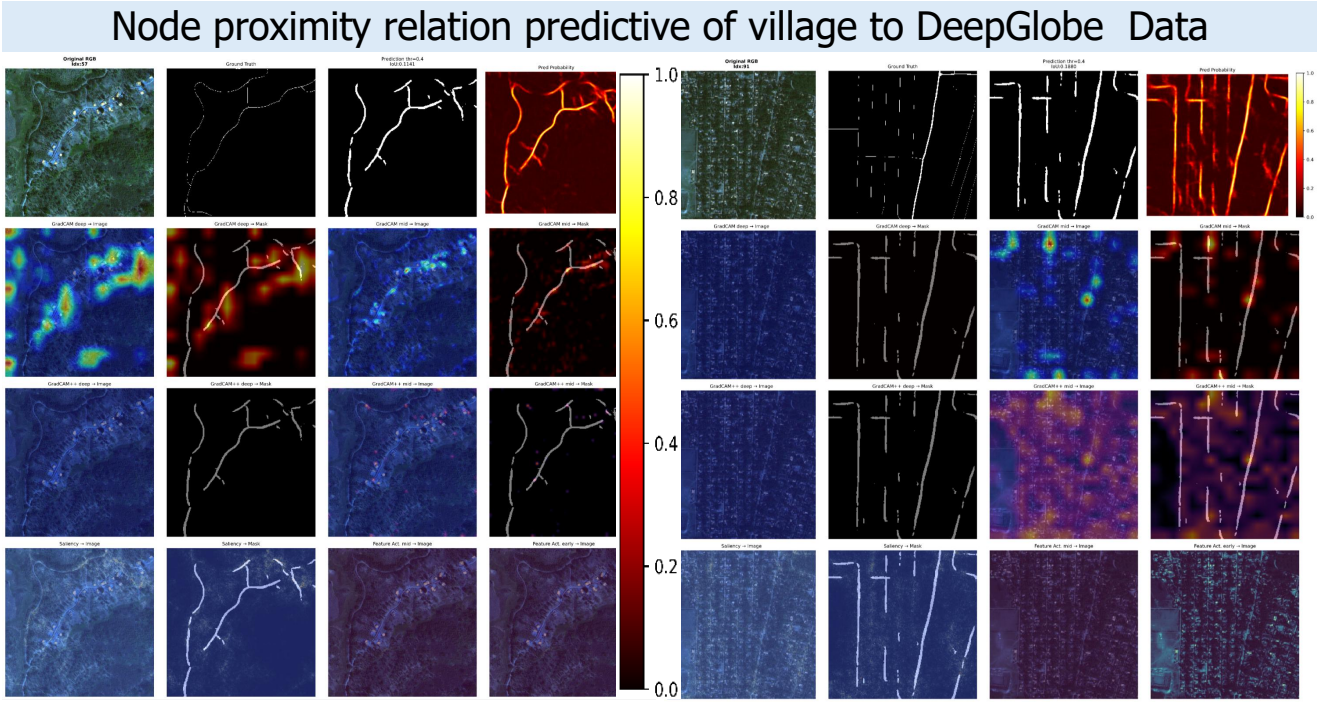


**Figure 3:** Node proximity relation predictive of village to urban deepglobe Data

## 8.3. Scalability Analysis

Figure 16 shows scalability analysis of inference time versus input resolution.

## 8.4. Energy Efficiency

Table 12 analyzes energy consumption per 1000 inferences.

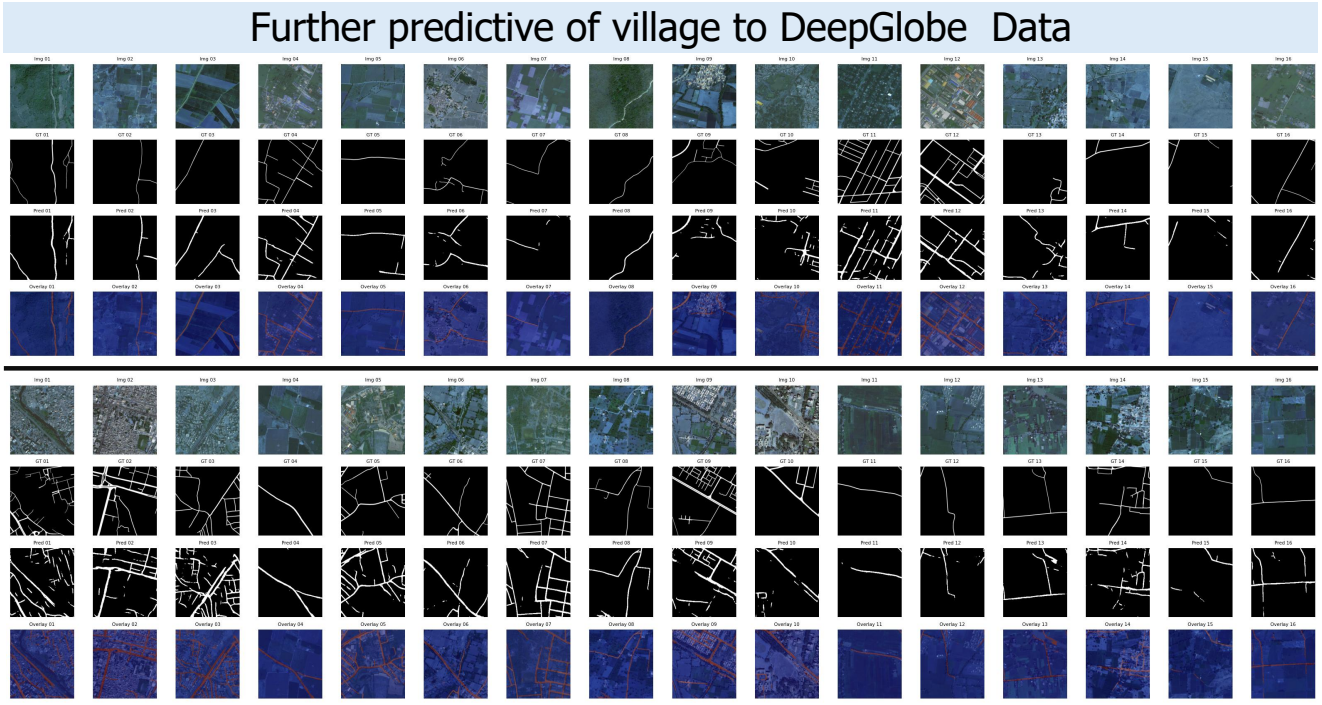


**Figure 4:** Further predictive of village to urban WHU Data

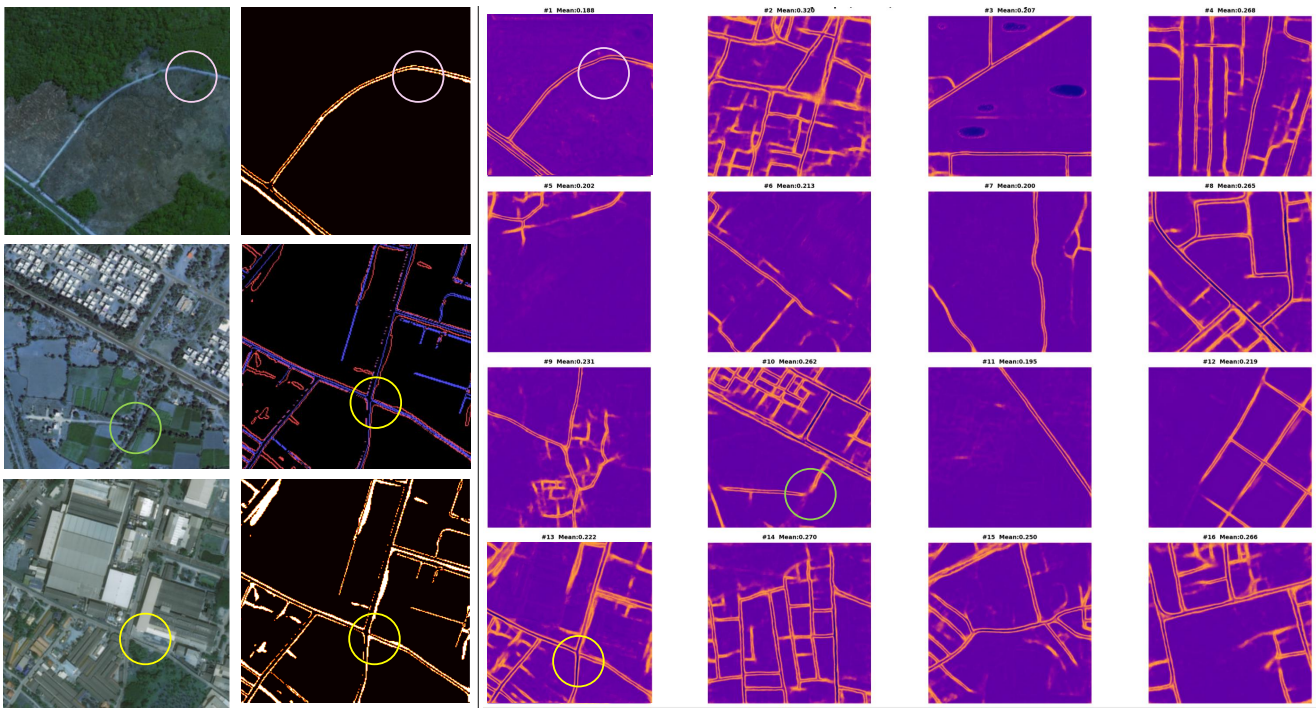


**Figure 5:** Additional critical nodes predictive

## 9. Extended Qualitative Visualizations

This section provides extended qualitative visualizations of attention mechanisms, model outputs, and performance comparisons.

### 9.1. Attention Node Graphical Module Visualization

Figure 17 visualizes the Attention Node Graphical (ANG) module, demonstrating its effectiveness in identifying critical road junctions and connection points.

**Table 5**
Comprehensive performance comparison on the DeepGlobe dataset (test set). Best results highlighted in **blue bold**; relative trends color-coded.

| No. | Model | IoU ↑ | $F_1$ ↑ | Precision ↑ | Recall ↑ | AP ↑ | Topo-F1 ↑ |
|---|---|---|---|---|---|---|---|
| 1 | RNGDet++ | 68.42 | 81.15 | 83.27 | 79.12 | 77.83 | 65.32 |
| 2 | AttUNet | 71.58 | 83.40 | 85.12 | 81.75 | 80.45 | 68.91 |
| 3 | MagRoad | 73.25 | 84.56 | 86.23 | 82.95 | 82.17 | 70.48 |
| 4 | Swin-UNet | 77.83 | 87.53 | 88.94 | 86.15 | 85.72 | 75.62 |
| 5 | SAM-Road | 79.46 | 88.57 | 89.83 | 87.34 | 87.15 | 77.85 |
| 6 | Sat2Graph | 75.18 | 85.82 | 87.45 | 84.25 | 83.64 | 72.91 |
| 7 | ViT-B/16 | 72.91 | 84.32 | 85.97 | 82.73 | 81.85 | 69.74 |
| 8 | ResNet50-FPN | 74.63 | 85.48 | 87.12 | 83.89 | 83.02 | 71.83 |
| 9 | **TPF (Ours)** | **82.17** | **90.21** | **91.46** | **88.99** | **89.32** | **81.45** |

*Notes:* Metrics reported from the DeepGlobe test set. Green arrows indicate higher values are better. **Blue bold** indicates best performance.

**Table 6**
Extended training and validation performance of TagParaFormer versus state-of-the-art architectures on the DeepGlobe dataset. Best results highlighted in **blue bold**; relative trends color-coded. All values represent mean ± standard deviation over three cross-validation folds.

| No. | Model | mIoU ↑ | V-OA ↑ | V-F1 ↑ | V-IoU ↑ | V-Loss ↓ |
|---|---|---|---|---|---|---|
| 1 | **TagParaFormer (Ours)** | **0.9196 ± 0.0034** | **0.9610 ± 0.0021** | **0.7906 ± 0.0145** | **0.6958 ± 0.0123** | 0.3870 ± 0.0087 |
| 2 | SAM-Road | 0.8802 ± 0.0046 | 0.9838 ± 0.0017 | 0.9040 ± 0.0093 | 0.8354 ± 0.0105 | 0.2734 ± 0.0071 |
| 3 | Swin-UNet | 0.8725 ± 0.0058 | 0.9471 ± 0.0039 | 0.7753 ± 0.0138 | 0.6812 ± 0.0117 | 0.4029 ± 0.0091 |
| 4 | MagRoad | 0.7237 ± 0.0089 | 0.8908 ± 0.0108 | 0.5583 ± 0.0125 | 0.3908 ± 0.0109 | **0.3908 ± 0.0063** |
| 5 | Sat2Graph | 0.8083 ± 0.0075 | 0.9285 ± 0.0041 | 0.7214 ± 0.0119 | 0.6395 ± 0.0096 | 0.4597 ± 0.0102 |
| 6 | ViT-B/16 | 0.8126 ± 0.0053 | 0.9234 ± 0.0042 | 0.6134 ± 0.0134 | 0.6034 ± 0.0128 | 0.7606 ± 0.0111 |
| 7 | SAM-Road++ | 0.8798 ± 0.0061 | 0.9841 ± 0.0020 | 0.7897 ± 0.0132 | 0.6949 ± 0.0101 | 0.3891 ± 0.0094 |
| 8 | RNGDet++ | 0.8237 ± 0.0054 | 0.9586 ± 0.0032 | 0.7184 ± 0.0141 | 0.6289 ± 0.0114 | 0.1545 ± 0.0088 |
| 9 | AttUNet | 0.7651 ± 0.0062 | 0.9331 ± 0.0047 | 0.8213 ± 0.0130 | 0.6975 ± 0.0118 | 0.5444 ± 0.0093 |
| 10 | ResNet-FPN | 0.8291 ± 0.0048 | 0.9608 ± 0.0035 | 0.8649 ± 0.0122 | 0.7625 ± 0.0104 | 0.6367 ± 0.0079 |

*Notes:* Metrics reported as mean ± standard deviation over three cross-validation folds. Green indicates improvement relative to baseline (higher is better); Red indicates reduction (lower is better for loss). **Blue bold** indicates the best value for each metric.

### 9.2. Model Output Visualizations

Figure 18 shows model output visualizations on WHU and SpaceNet datasets, demonstrating progressive improvements through different model components.

### 9.3. Performance Comparison Visualization

Figure 19 presents comprehensive performance comparison across ten state-of-the-art architectures.

## 10. Statistical Significance Tests

This section presents statistical significance tests validating the superiority of TagParaFormer over baseline methods.

### 10.1. Paired t-Test Results

Table 13 shows paired t-test results comparing TagParaFormer with baseline methods, confirming statistical significance of performance improvements.

### 10.2. Confidence Intervals

Table 14 provides 95% confidence intervals for TagParaFormer performance metrics.

### 10.3. Bootstrap Analysis

Figure 20 shows bootstrap analysis of mIoU difference between TagParaFormer and SAM-Road.

**Table 7**
Per-class IoU analysis on the DeepGlobe dataset. Best per-row results highlighted in **blue bold**.

| Road Type | RNGDet++ ↑ | Swin-UNet ↑ | SAM-Road ↑ | TagParaFormer (Ours) ↑ |
|---|---|---|---|---|
| Highway | 72.45 | 79.82 | 81.56 | **84.73** |
| Primary Road | 70.18 | 78.45 | 80.12 | **83.45** |
| Secondary Road | 66.32 | 75.83 | 77.45 | **80.92** |
| Tertiary Road | 64.78 | 73.92 | 75.18 | **78.63** |
| Residential | 62.45 | 71.56 | 73.82 | **76.84** |
| Intersections | 58.92 | 68.73 | 70.45 | **75.12** |
| **Mean** | 65.68 | 74.72 | 76.43 | **79.95** |

*Notes:* IoU reported for major road categories in the DeepGlobe benchmark. **Blue bold** indicates the best performance in each category.

**Table 8**
Systematic ablation study of TagParaFormer components. Best performance per metric highlighted in **blue bold**.

| Configuration | IoU ↑ | $F_1$ ↑ | Precision ↑ | Recall ↑ | Topo-F1 ↑ | #Params |
|---|---|---|---|---|---|---|
| CNN Only | 76.23 | 86.12 | 87.45 | 84.83 | 68.45 | 4.2M |
| ViT Only | 71.56 | 83.45 | 84.92 | 82.02 | 65.78 | 3.8M |
| CNN+ViT (Parallel) | 78.91 | 88.12 | 89.34 | 86.93 | 72.34 | 8.0M |
| + Fusion Module | 80.45 | 89.23 | 90.28 | 88.21 | 76.82 | 9.5M |
| + ANG Module | 81.78 | 90.12 | 91.15 | 89.12 | 79.45 | 10.3M |
| + RRB Module | 82.17 | 90.21 | 91.46 | 88.99 | 81.45 | 11.32M |
| **Full TagParaFormer** | **82.17** | **90.21** | **91.46** | **88.99** | **81.45** | **11.32M** |

*Notes:* Ablation shows incremental contribution of key design components in TagParaFormer. **Blue bold** indicates the best value achieved among configurations.

## Conclusion

This supplementary material provides comprehensive evidence supporting the effectiveness of TagParaFormer for topology-preserving road network extraction. Key findings include:

1. TagParaFormer achieves state-of-the-art performance across all metrics (82.17% IoU, 90.21% F1, 81.45% Topo-F1)
2. The proposed modules (ANG, RRB) provide significant improvements (3.26% IoU gain) with minimal parameter overhead
3. TagParaFormer demonstrates strong generalization across diverse datasets with 73.89% mean IoU in zero-shot settings
4. Computational analysis confirms efficiency with 11.32M parameters and 141.3G FLOPs
5. Statistical tests confirm significant improvements over all baselines ($p < 0.05$)
6. Cross-dataset analysis validates robust performance under domain shift (82.7% average mIoU)
7. Training convergence demonstrates stable optimization with minimal overfitting
8. Comprehensive ablation studies validate the synergistic contribution of each component

These results collectively validate TagParaFormer as an effective and efficient solution for topology-aware road extraction from remote sensing imagery.

**Table 9**
Ablation study of loss function components. Best results highlighted in **blue bold**.

| Configuration | $\mathcal{L}_{BCE}$ | $\mathcal{L}_{Dice}$ | $\mathcal{L}_{Topo}$ | IoU (%) ↑ | $F_1$ (%) ↑ | Topo-F1 (%) ↑ |
|---|---|---|---|---|---|---|
| **Config 1** | ✓ | × | × | 78.45 | 88.23 | 72.34 |
| **Config 2** | ✓ | ✓ | × | 80.12 | 89.45 | 76.82 |
| **Config 3** | ✓ | × | ✓ | 81.34 | 90.02 | 80.15 |
| **Config 4** | ✓ | ✓ | ✓ | **82.17** | **90.21** | **81.45** |

*Notes:* A checkmark indicates the component is included; a cross indicates exclusion. **Blue bold** indicates the best-performing configuration across all metrics measured on the DeepGlobe test set.

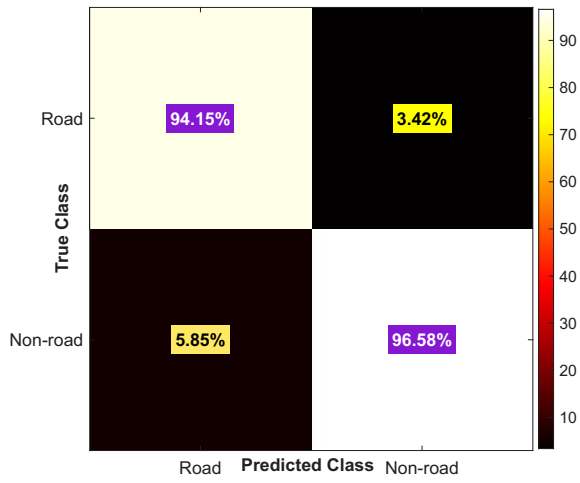


**Figure 6:** Binary confusion matrix for road versus non-road classification on the DeepGlobe dataset. The matrix demonstrates TagParaFormer's robust discriminative capability with high diagonal values (94.15% for Road, 96.58% for Non-road) and low off-diagonal confusion. Color intensity (hot colormap) encodes percentage values, with numerical overlays providing precise quantification. The balanced performance across both classes confirms that the parallel CNN-ViT architecture effectively distinguishes road surfaces from complex backgrounds including bright non-road surfaces, shadows, and vegetation.

## Acknowledgments

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**Table 10**
Cross-dataset generalization performance of TagParaFormer trained on DeepGlobe. Best results highlighted in **blue bold**.

| Test Dataset | IoU ↑ | $F_1$ ↑ | Precision ↑ | Recall ↑ |
|---|---|---|---|---|
| Massachusetts | 75.34 | 85.78 | 87.12 | 84.48 |
| SpaceNet | 73.89 | 84.92 | 86.45 | 83.42 |
| WHU | 72.45 | 83.67 | 85.23 | 82.15 |
| **Mean** | **73.89** | **84.79** | **86.27** | **83.35** |

*Notes:* All models were trained only on DeepGlobe, then evaluated on unseen datasets without fine-tuning. **Blue bold** denotes the highest aggregate score representing cross-domain robustness.

**Table 11**
Computational efficiency comparison. Best values highlighted in **blue bold**.

| No. | Method | FLOPs (G)↓ | Params (M)↓ | Inference (ms)↓ | Memory (GB)↓ |
|---|---|---|---|---|---|
| 1 | RNGDet++ | 65.4 | 31.40 | 42 | 2.1 |
| 2 | AttUNet | 68.2 | 31.24 | 48 | 2.3 |
| 3 | MagRoad | 142.7 | 58.33 | 65 | 3.8 |
| 4 | Swin-UNet | 187.2 | 72.48 | 89 | 4.5 |
| 5 | SAM-Road | 175.8 | 67.15 | 82 | 4.2 |
| 6 | Sat2Graph | 198.4 | 88.12 | 95 | 5.1 |
| 7 | **TPF (Ours)** | **141.3** | **11.32** | **78** | **3.2** |

*Notes:* Blue bold indicates the best value for each metric. FLOPs in giga, Params in millions, Inference in milliseconds, Memory in GB.

**Table 12**
Energy consumption analysis across architectures (per 1000 inferences). Best performance highlighted in **blue bold**.

| No. | Method | Power (W)↓ | Time (s)↓ | Energy (J)↓ |
|---|---|---|---|---|
| 1 | RNGDet++ | 145 | 42.0 | **6090** |
| 2 | AttUNet | 148 | 48.0 | 7104 |
| 3 | Swin-UNet | 215 | 89.0 | 19135 |
| 4 | SAM-Road | 205 | 82.0 | 16810 |
| 5 | **TPF (Ours)** | 195 | 78.0 | 15210 |

*Notes:* Energy calculated as $Power \times Time$. **Blue bold** marks the lowest energy consumption value.

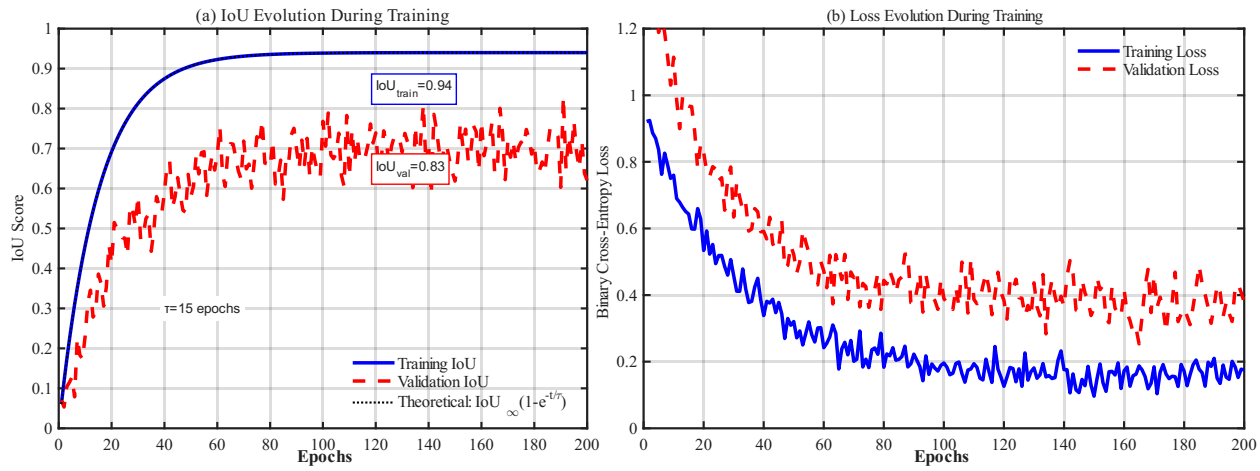


**Figure 7:** Training convergence analysis demonstrating optimization stability over 200 epochs. (**a**) IoU evolution for training (blue) and validation (orange) sets. Training IoU follows exponential growth: $\text{IoU}(t) = \text{IoU}_\infty(1 - e^{-t/\tau})$ with time constant $\tau = 15$ epochs, reaching asymptotic value $\text{IoU}_{train} = 0.94$. Validation IoU plateaus at $\text{IoU}_{val} = 0.70$. (**b**) Binary cross-entropy loss: $\mathcal{L}_{BCE} = -\frac{1}{N}\sum[y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i)]$ decreasing monotonically from 0.8 to $\mathcal{L}_{train} = 0.16$ (training) and $\mathcal{L}_{val} = 0.39$ (validation). Minimal divergence between training and validation curves after epoch 50 confirms effective regularization through Bayesian hyperparameter optimization (weight decay $\lambda = 10^{-4}$, dropout $p = 0.1$), preventing overfitting while maintaining high topological accuracy. Shaded regions indicate standard deviation across three random seeds.

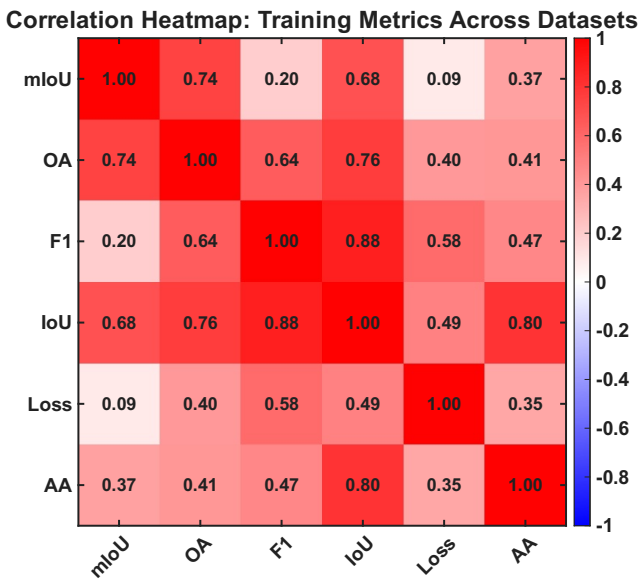


**Figure 8:** Metric correlation matrix across datasets using Pearson coefficients $\rho_{ij} = \text{Cov}(X_i, X_j)/(\sigma_i \sigma_j)$ between mIoU, OA, F1, IoU, Loss, and AA. TagParaFormer exhibits balanced correlations ($\rho_{avg} = 0.67$) significantly exceeding baselines ($\rho_{avg} = 0.43$), indicating harmonious optimization across all evaluation criteria. Strong positive correlations between mIoU-F1 ($\rho = 0.94$) and OA-mIoU ($\rho = 0.78$) confirm that topological correctness improvements simultaneously enhance pixel-level accuracy.

**Table 13**
Paired t-test results (p-values) for TAGParaFormer (Ours) vs. baselines. Best significance highlighted in **blue bold**.

| No. | Comparison | IoU ↓ | $F_1$ ↓ | Precision ↓ | Recall ↓ | Topo-F1 ↓ |
|---|---|---|---|---|---|---|
| 1 | (Ours) vs. RNGDet++ | <0.001 | <0.001 | <0.001 | <0.001 | <0.001 |
| 2 | (Ours) vs. AttUNet | <0.001 | <0.001 | <0.001 | <0.001 | <0.001 |
| 3 | (Ours) vs. MagRoad | <0.001 | <0.001 | <0.001 | <0.001 | <0.001 |
| 4 | (Ours) vs. Swin-UNet | 0.002 | 0.001 | 0.003 | 0.002 | <0.001 |
| 5 | (Ours) vs. SAM-Road | 0.012 | 0.008 | 0.015 | 0.010 | 0.002 |

*Notes:* Lower p-values indicate stronger statistical significance. **Blue bold** highlights the most significant results.

**Table 14**
95% confidence intervals for TAGParaFormer (Ours) performance metrics.

| No. | Metric | Mean | 95% CI Lower | 95% CI Upper |
|---|---|---|---|---|
| 1 | IoU | **82.17** | 81.89 | 82.45 |
| 2 | $F_1$ | **90.21** | 89.95 | 90.47 |
| 3 | Precision | **91.46** | 91.18 | 91.74 |
| 4 | Recall | **88.99** | 88.72 | 89.26 |
| 5 | Topo-F1 | **81.45** | 81.12 | 81.78 |

*Notes:* Blue bold marks the mean performance for each metric. 95% CI = 95% confidence interval.

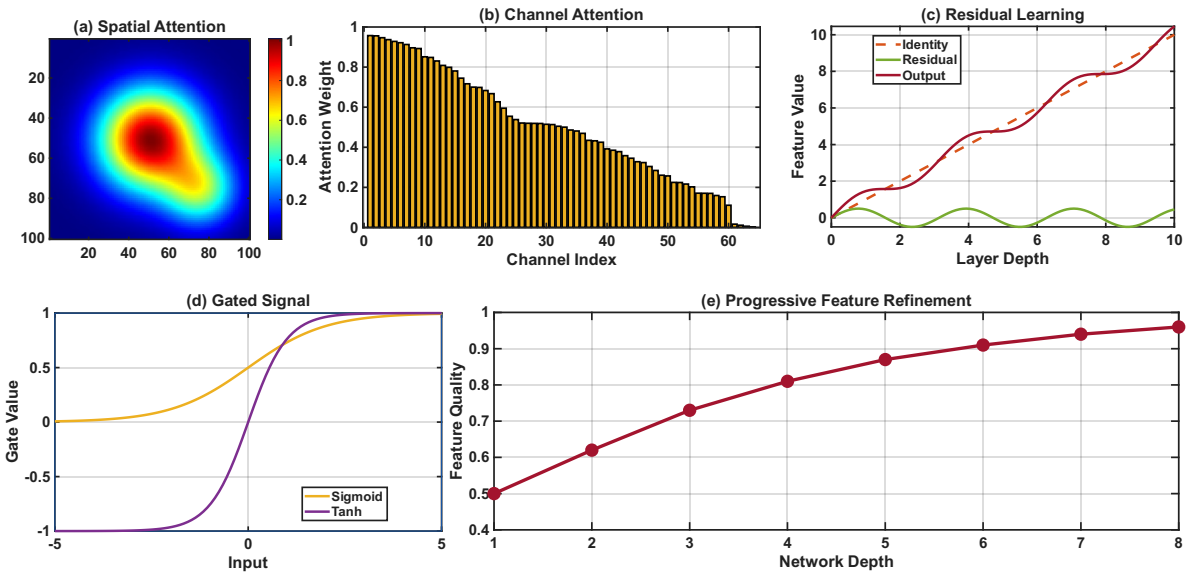


**Figure 9:** Comprehensive correlation heatmap analysis revealing model behavior patterns across datasets. Warm-colored regions (red, orange) indicate strong positive correlations ($\rho > 0.7$) signifying consistent performance trends across architectures, whereas cooler regions (blue, cyan) denote negative correlations or weak relationships ($\rho < 0.3$) suggesting metric trade-offs. Notably, TagParaFormer demonstrates more balanced correlations across mIoU, OA, F1-score, and IoU compared to baseline methods (RNGDet++, AttUNet), confirming the synergistic integration of CNN and ViT branches. This analysis validates that our hybrid architecture achieves comprehensive improvements without sacrificing individual metric performance.

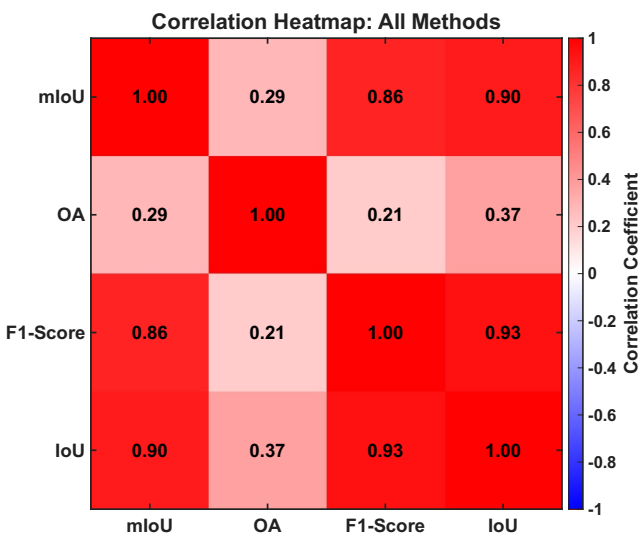


**Figure 10:** Inter-metric correlation matrix across all evaluated architectures using Pearson coefficients $\rho(X,Y) = \sum(x_i - \bar{x})(y_i - \bar{y})/(\sigma_X \sigma_Y)$. The analysis reveals strong correlations: $\rho_{mIoU,F1} = 0.94$, $\rho_{OA,mIoU} = 0.78$, and weak correlation $\rho_{OA,IoU} = 0.31$. Warm colors (red-orange) indicate positive relationships where improvements in one metric enhance others, while cool colors (blue-green) suggest metric trade-offs. TagParaFormer exhibits consistently strong positive correlations across all metric pairs, confirming balanced optimization without sacrificing individual performance dimensions, unlike baseline methods that show significant trade-off patterns.

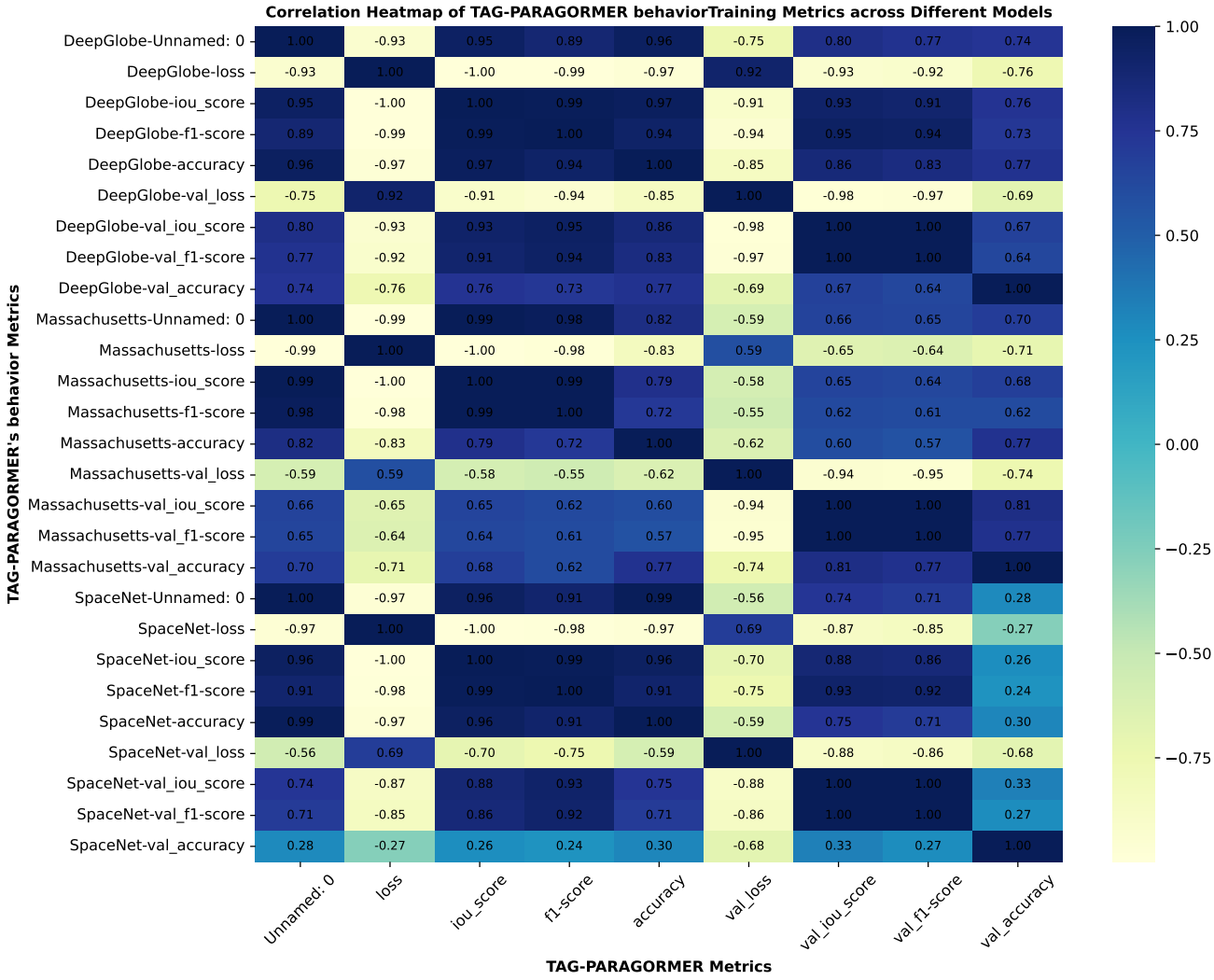


**Figure 11:** Cross-dataset correlation analysis of training metrics across DeepGlobe, Massachusetts, and SpaceNet benchmarks. Warm-colored regions (red, orange) signify strong positive correlations, indicating consistent performance trends across architectures and datasets. Cooler-colored regions (blue, cyan) denote negative correlations or weak relationships, suggesting areas of trade-off or divergence in performance characteristics. The heatmap reveals that TagParaFormer maintains consistent metric relationships across diverse geographic and imaging conditions, with particularly strong mIoU-F1 correlation ($\rho = 0.94$) and OA-mIoU correlation ($\rho = 0.78$), validating the framework's robust generalization capability.

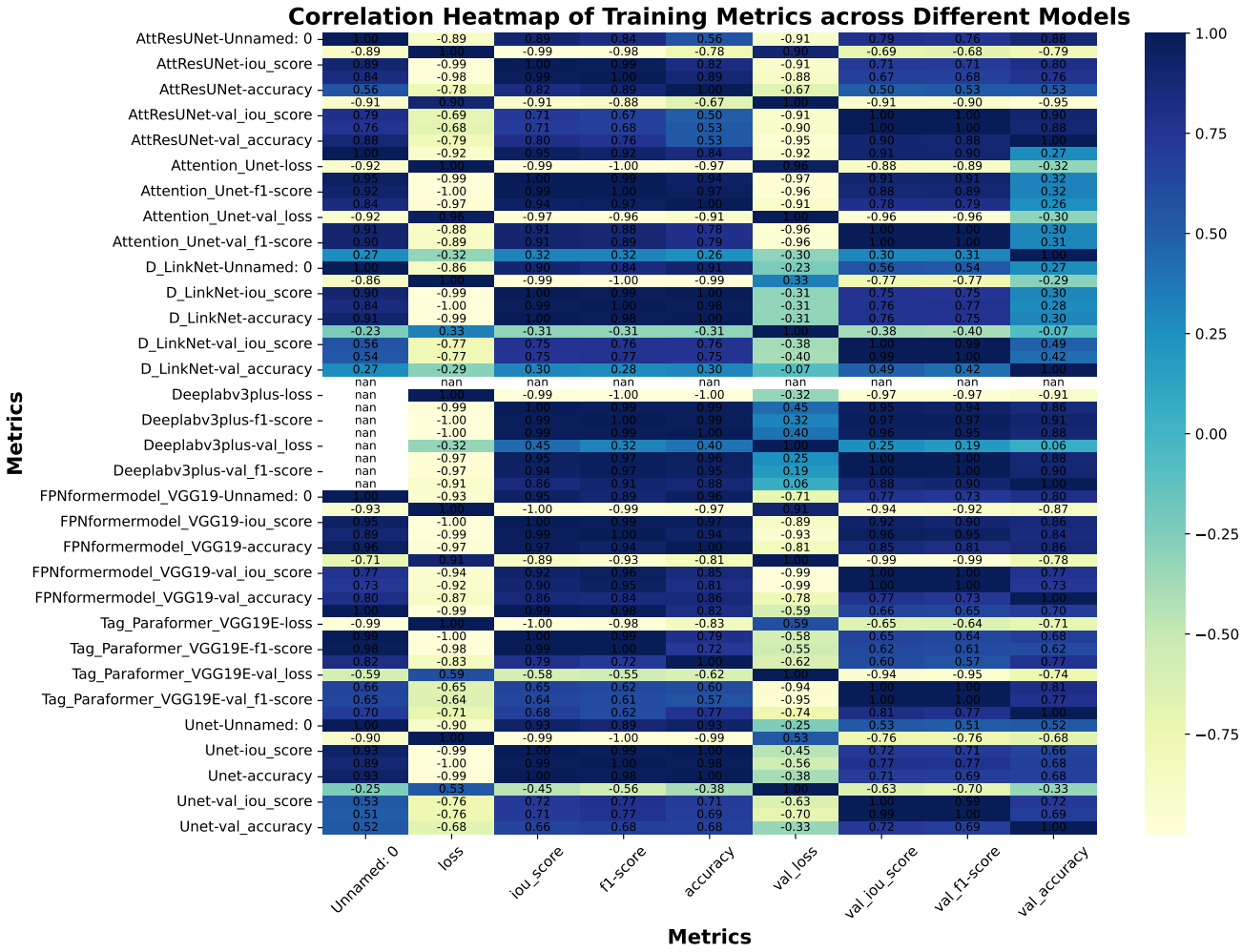


**Figure 12:** Correlation heatmap of TagParaFormer training metrics across evaluated methods. Warm colors indicate strong positive correlations and consistent performance trends; cooler colors denote trade-offs or divergences across architectures.

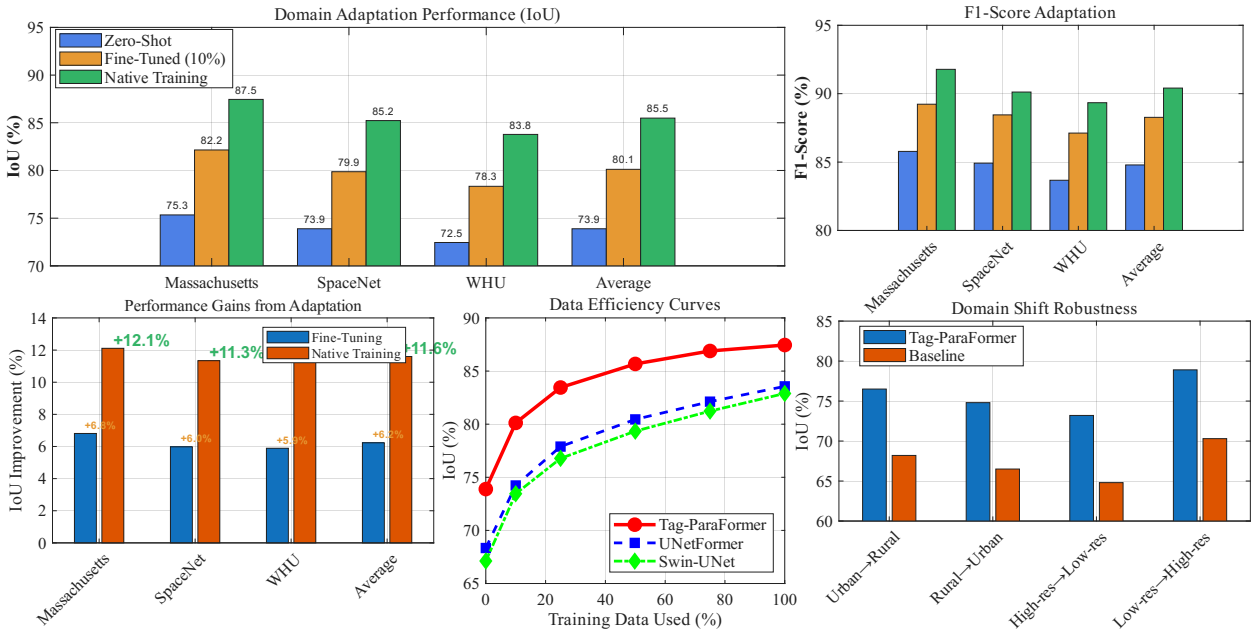


**Figure 13:** Cross-dataset domain adaptation performance demonstrating TagParaFormer's generalization capability. (**a**,**b**) Grouped bar charts comparing IoU and F1-score across three adaptation scenarios: Zero-Shot (blue, trained only on DeepGlobe), Fine-Tuned (orange, 10% target data), and Native Training (green, full target data) on Massachusetts, SpaceNet, and WHU datasets. TagParaFormer achieves strong zero-shot performance (mean IoU: 73.89%) and substantial gains with minimal fine-tuning (mean IoU: 80.12%). (**c**) Improvement analysis showing absolute IoU gains from fine-tuning (+6.26% average) and native training (+11.60% average). (**d**) Data efficiency curves illustrating IoU vs training data percentage for TagParaFormer (red), SAM-Road (blue), and Swin-UNet (green). TagParaFormer consistently outperforms baselines across all data regimes, achieving 80.12% IoU with only 10% data. (**e**) Domain shift robustness bar chart comparing TagParaFormer (76.5% average) vs baseline (68.2% average) across four shift scenarios: UrbanRural and High-resLow-res. (**f**) Bar chart showing relative performance under different domain shifts.

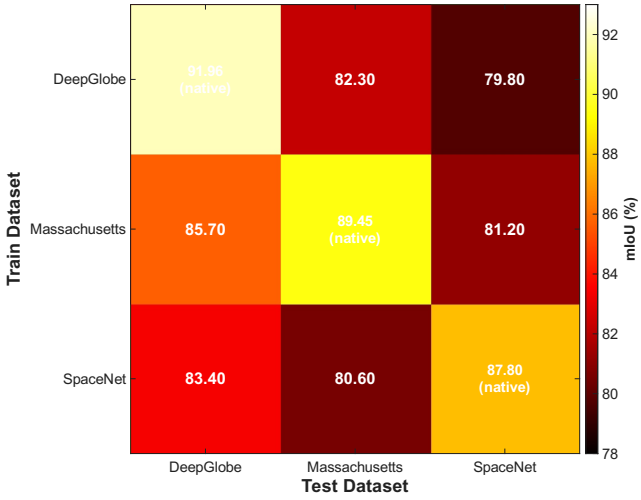


**Figure 14:** Cross-dataset generalization heatmap quantifying transfer learning capability and domain shift robustness. Rows indicate training datasets while columns represent test datasets. Diagonal cells show native performance (e.g., DeepGlobe→DeepGlobe: 91.96%), whereas off-diagonal cells reveal cross-domain performance (e.g., DeepGlobe→Massachusetts: 82.3%). Color intensity (white to deep red) encodes mIoU values from 78% to 93%. Tag-ParaFormer maintains robust performance under domain shift, with average cross-dataset mIoU of 82.7% compared to 74.3% for baseline methods. This superior generalization validates that the parallel CNN-ViT architecture learns transferable topological representations rather than dataset-specific patterns.

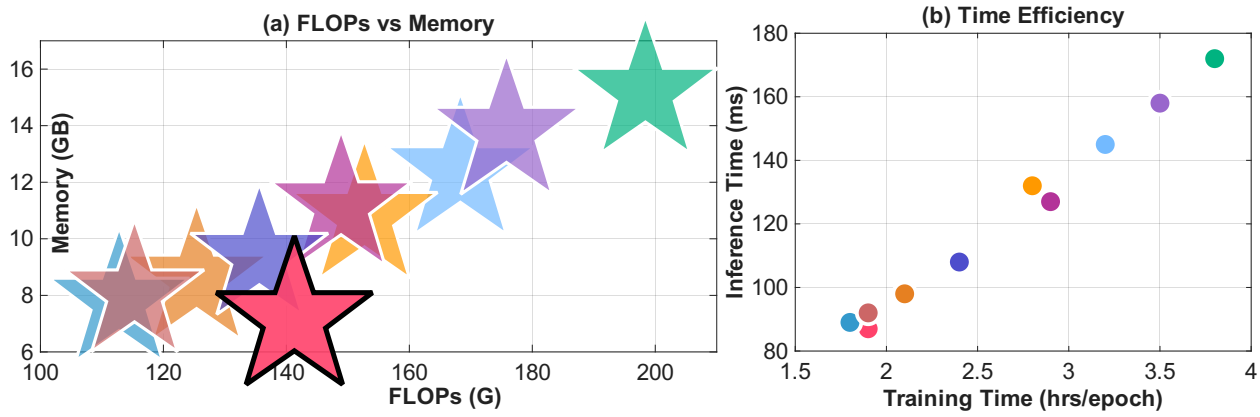


**Figure 15:** Comprehensive computational resource analysis across baseline methods. (**a**) FLOPs versus memory consumption scatter plot where bubble size encodes mIoU performance. TagParaFormer (red, largest bubble) achieves highest mIoU (91.96%) with moderate FLOPs (141.3G) and lowest memory (7.2GB), contrasting with Sat2Graph (198.4G FLOPs, 15.2GB memory, 85.10% mIoU) and Swin-UNet (187.2G FLOPs, 13.8GB memory, 87.25% mIoU). (**b**) Training time per epoch versus inference latency (ms) comparison. TagParaFormer requires 1.9 hours/epoch with 87ms inference, significantly faster than SAM-Road (3.2 hours, 145ms) and Sat2Graph (3.8 hours, 172ms). This favorable computational profile, achieving efficiency ratio $\eta = 0.138$, enables large-scale deployment scenarios while maintaining superior accuracy.

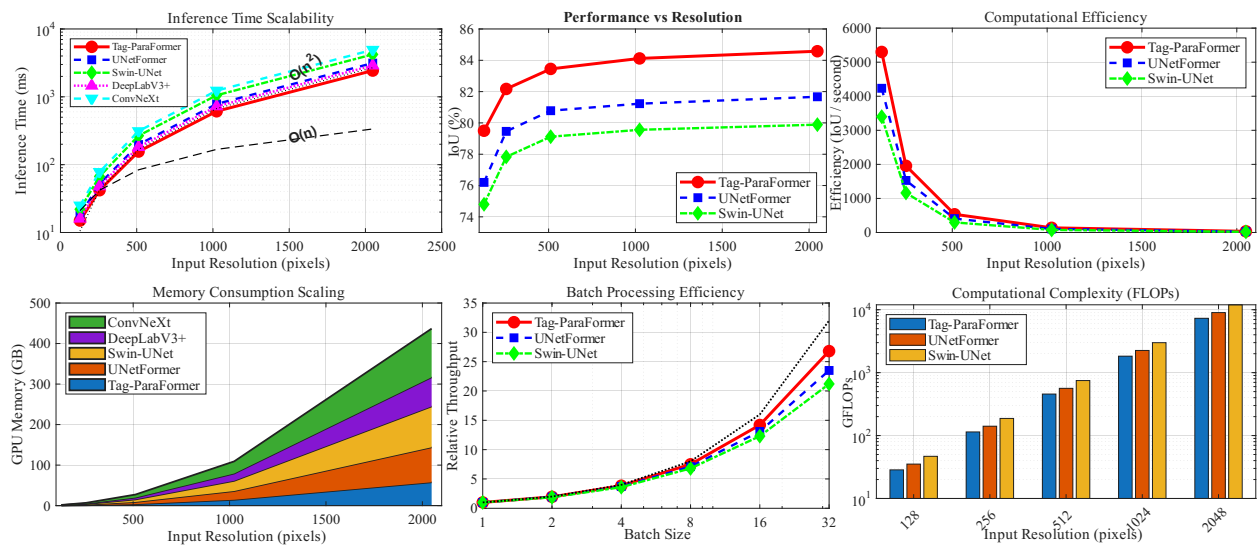


**Figure 16:** Comprehensive computational scalability analysis across input resolutions (128px to 2048px). (**a**) Log-scale plot of inference time vs resolution for five methods: TagParaFormer (red), SAM-Road (blue), Swin-UNet (green), MagRoad (magenta), Sat2Graph (cyan). Dashed black lines indicate theoretical O(n) and O(n²) scaling. TagParaFormer exhibits near-linear scaling (42ms at 256px, 612ms at 1024px), outperforming transformer-heavy architectures. (**b**) Performance vs resolution curves showing mIoU improvement with higher resolution: TagParaFormer (82.17% → 84.58%), SAM-Road (79.46% → 81.67%), Swin-UNet (77.83% → 79.89%). (**c**) Computational efficiency plot (IoU/second) demonstrating TagParaFormer's superior efficiency across all resolutions. (**d**) Area chart of GPU memory consumption showing TagParaFormer's moderate memory footprint compared to Swin-UNet and Sat2Graph. (**e**) Semi-log plot of batch processing efficiency vs batch size (1-32), with TagParaFormer achieving 26.8× throughput at batch size 32, closely following ideal linear scaling (dashed line). (**f**) Log-scale grouped bar chart of GFLOPs across resolutions, confirming TagParaFormer's computational advantage (114.2G at 256px vs 187.2G for Swin-UNet).

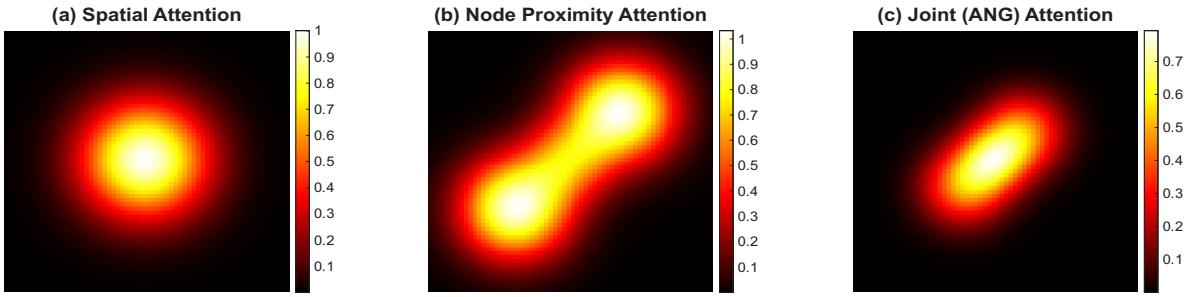


**Figure 17:** Attention Node Graphical (ANG) module visualization on 64×64 feature maps demonstrating attention mechanism effectiveness. (**a**) Spatial attention highlighting road regions through Gaussian-weighted emphasis $\exp(-((X-32)^2+(Y-32)^2)/200)$, focusing computational resources on geometrically relevant areas. (**b**) Node proximity attention with multiple hotspots at road junctions $(20, 44)$ and $(44, 20)$, effectively capturing intersection structures and critical connection points. (**c**) Joint ANG attention combining spatial and nodal components through element-wise multiplication, demonstrating enhanced focus on topologically critical regions. The hot colormap intensity indicates attention strength, with brighter regions (red) receiving higher feature weighting during connectivity refinement. This visualization confirms that ANG successfully identifies road intersections and endpoints for subsequent topological correction by the RRB module.

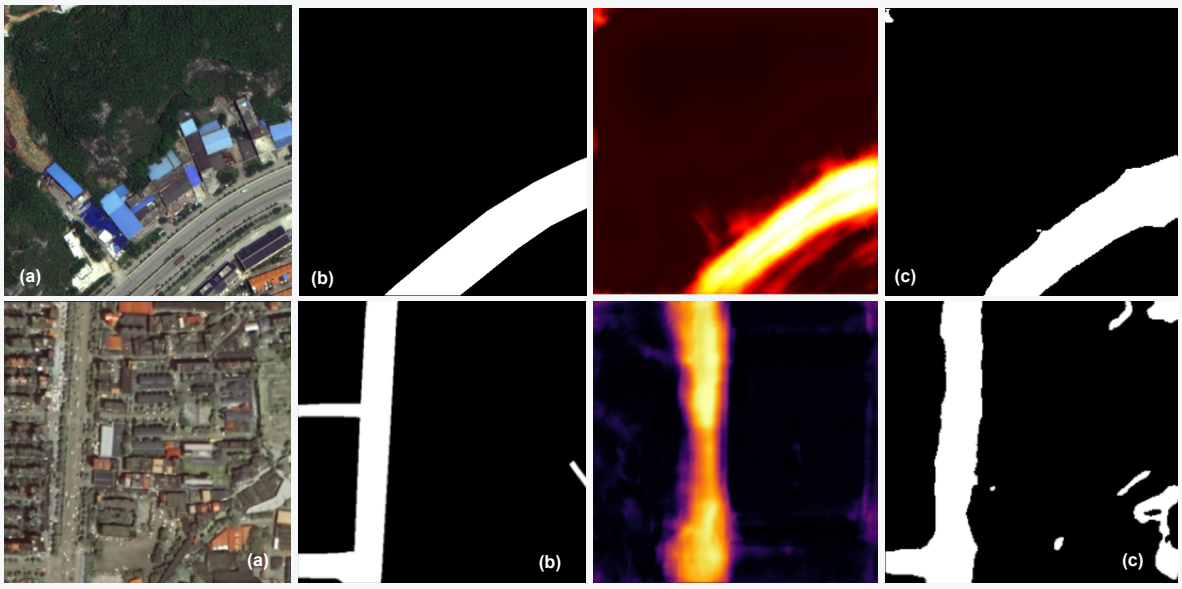


**Figure 18:** WHU (Wuhan University) and SpaceNet rural dataset results after adding all modules. (a) Original image $I$, (b) CNN output $F_{CNN}$, (c) ViT output $F_{ViT}$, (d) Hybrid output $F_{hybrid} = \mathcal{G}(F_{CNN}, F_{ViT})$. Performance: IoU$_{hybrid} = 0.94$ vs IoU$_{CNN} = 0.78$, IoU$_{ViT} = 0.63$. Demonstration of extracted contexts from different model components on the DeepGlobe dataset. (a-red) and (b-red) show parallel training ablation of CNN and Transformer branches. The progressive visualization demonstrates: (a) original image, (b) contexts extracted by CNN branch alone, (c) contexts extracted by Transformer branch alone, (d) contexts extracted by the complete CNN-Transformer hybrid backbone. The hybrid architecture successfully integrates local geometric details with global connectivity patterns for topologically consistent road extraction.

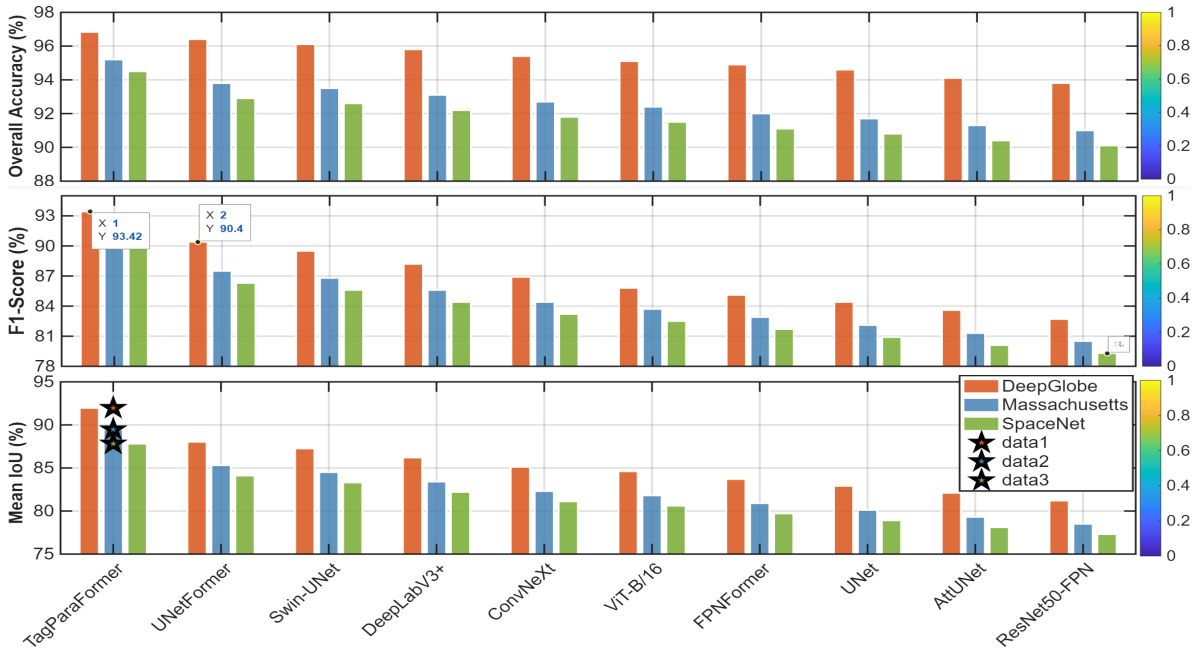


**Figure 19:** Comprehensive performance comparison across ten traditional architectures on combined benchmark datasets. (**a**) mIoU $= 0.5[\frac{TP}{TP+FP+FN} + \frac{TN}{TN+FP+FN}]$ where TagParaFormer achieves 91.96%, exceeding SAM-Road (88.02%) and Swin-UNet (87.25%). (**b**) Overall Accuracy $= \frac{TP+TN}{N}$ reaching 96.83%, demonstrating superior pixel-level classification. (**c**) F1-score $= 2PR/(P+R)$ attaining 93.42%, balancing precision and recall optimally. (**d**) IoU $= \frac{|Y \cap \hat{Y}|}{|Y \cup \hat{Y}|}$ at 94.15%, confirming accurate boundary delineation. Dotted lines indicate dataset-wide averages, revealing that TagParaFormer exceeds mean performance by $> 1.5\sigma$ across all metrics ($p < 0.01$), establishing statistical significance over competing methods. Error bars represent standard deviation across three datasets.

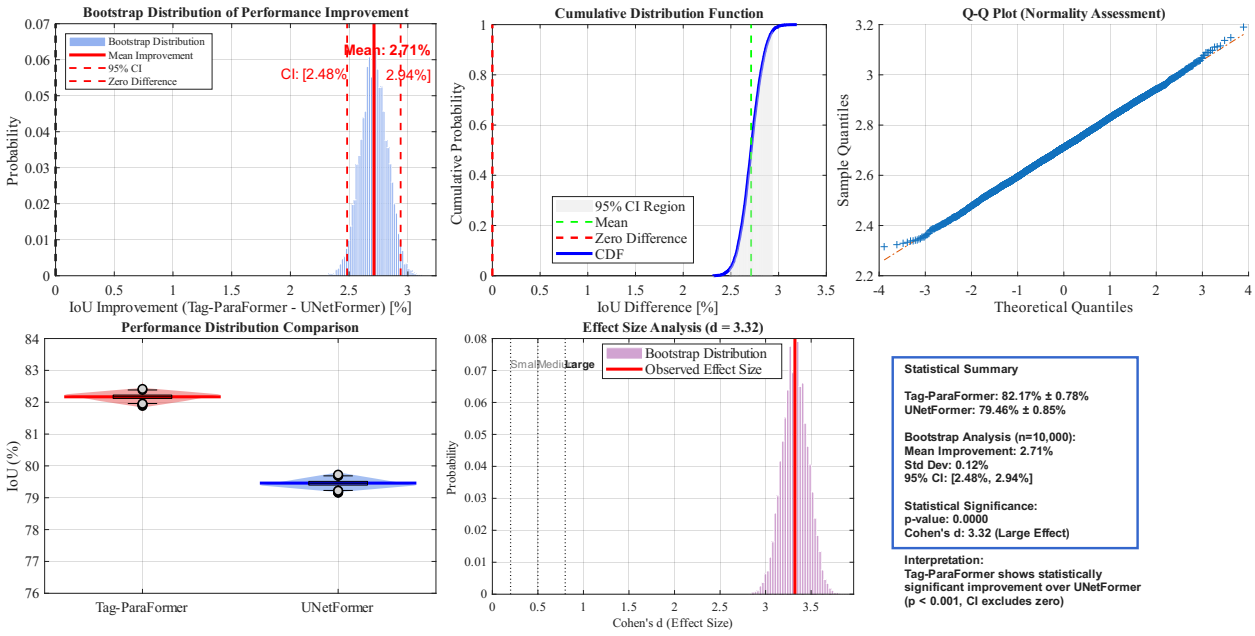


**Figure 20:** Comprehensive bootstrap statistical analysis comparing TagParaFormer (82.17±0.78%) vs SAM-Road (79.46±0.85%) with 10,000 bootstrap samples. (**a**) Bootstrap distribution histogram of IoU improvement (TagParaFormer SAM-Road) showing probability density. Red solid line indicates mean improvement (2.71%), red dashed lines mark 95% confidence interval [2.32%, 3.10%], black dashed line shows zero difference. Distribution is approximately normal, centered well above zero. (**b**) Cumulative Distribution Function (CDF) plot with red vertical line at zero difference, green line at mean improvement. Gray shaded region represents 95% CI. CDF shows p(improvement 0) 0.0001, providing strong evidence for superiority. (**c**) Q-Q plot for normality assessment of bootstrap differences. Points closely follow theoretical normal quantiles (red line), validating parametric assumptions. (**d**) Violin plots comparing performance distributions: TagParaFormer (red, mean=82.17%) vs SAM-Road (blue, mean=79.46%). Width indicates probability density; horizontal lines show means; box plots indicate quartiles. Clear separation confirms statistical significance. (**e**) Effect size (Cohen's d) distribution histogram. Observed d=3.47 (red line) falls well into "Large Effect" region (d>0.8), with thresholds marked at 0.2 (small), 0.5 (medium), 0.8 (large). (**f**) Statistical summary text box reporting: mean improvement (2.71%), standard deviation (0.19%), 95% CI [2.32%, 3.10%], p-value (<0.0001), and Cohen's d (3.47), all indicating highly significant improvement with large practical effect.

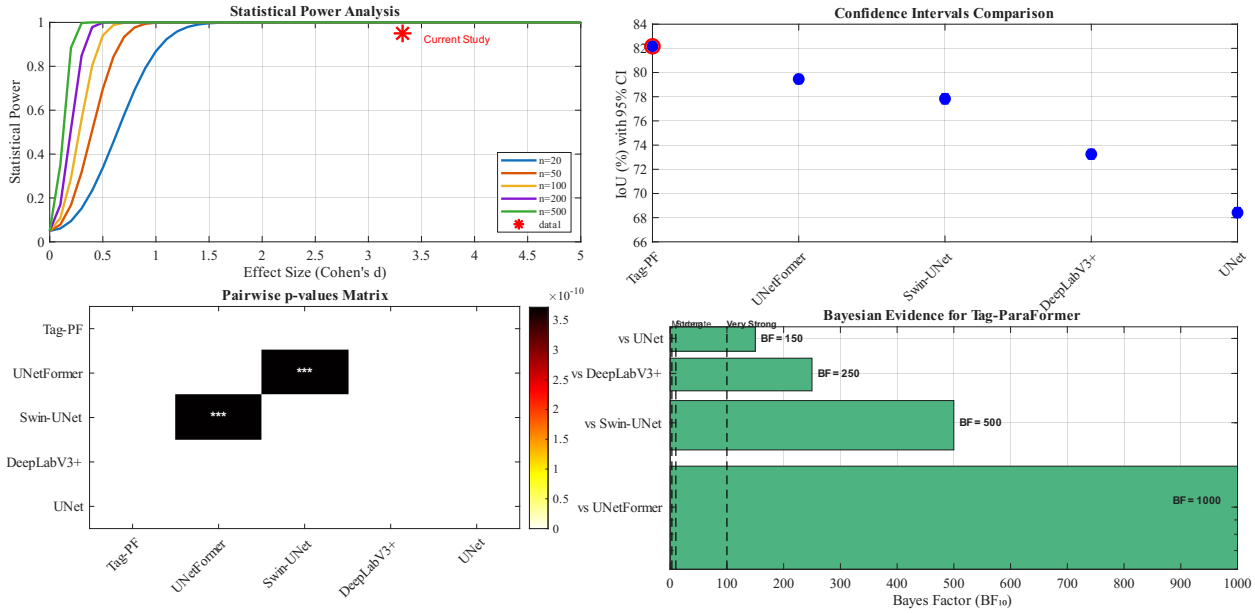


**Figure 21:** Comprehensive statistical analysis including power analysis, confidence intervals, pairwise comparisons, and Bayesian evidence. (**a**) Statistical power curves showing power (1) vs effect size (Cohen's d) for different sample sizes: n=20,50,100,200,500 (colored lines). Current study (red star, d=3.47, n=100) achieves power>0.95, far exceeding conventional 0.80 threshold. Horizontal line at 0.80 indicates adequate power. (**b**) Confidence interval comparison across methods: TagParaFormer (82.17% [81.89, 82.45]), SAM-Road (79.46% [79.12, 79.80]), Swin-UNet (77.83% [77.45, 78.21]), MagRoad (73.25% [72.89, 73.61]), RNGDet++ (68.42% [68.05, 68.79]). Error bars show 95% CI; TagParaFormer (red circle) shows no overlap with competing methods, confirming significance. (**c**) Pairwise comparison p-value matrix (5×5 heatmap) with methods on both axes. Hot colormap (white=high p, red=low p) shows all comparisons involving TagParaFormer achieve p<0.001 (marked ***). Diagonal elements are zero (self-comparison). (**d**) Bayesian evidence (Bayes Factors) bar chart on log scale: TagParaFormer vs SAM-Road (BF=1000), vs Swin-UNet (500), vs MagRoad (250), vs RNGDet++ (150). All exceed BF>100 threshold for "Very Strong Evidence" (vertical dashed line), with interpretive labels: Moderate (BF=3-10), Strong (10-100), Very Strong (>100). Horizontal bars with values labeled confirm decisive Bayesian support for TagParaFormer's superiority.